Embryo Scan: Automated Morphokinetic Annotation, Viability Ranking and Gardner Grading of Human IVF Embryos

Technical report · Cody Huang · October 2026

Code and data splits: github.com/cha205/embryo-scan · Interactive demonstrator: embryo-scan.vercel.app

> Research software. The models described here have not been validated for clinical use and must not be used to make treatment decisions.

Abstract

Background. Embryo selection in in vitro fertilization (IVF) relies on visual assessment of embryo morphology and, where time-lapse imaging is available, on manually annotated division timings. Both are labour-intensive and subject to inter-observer variability, and published selection algorithms often transfer poorly between clinics.

Methods. An open, end-to-end pipeline was developed and evaluated on two public single-centre datasets. (i) A ResNet-34 classifier assigns each frame of a time-lapse recording to one of 16 developmental phases using a three-frame temporal context; a dynamic-programming decoder enforces monotonic progression and yields division times (704 videos, 542 patients). The deep model was compared with a classical image-processing baseline implemented in MATLAB and scored with identical rules. (ii) A logistic regression on division times observed up to 72 h ranks embryos by the probability of blastocyst formation by 116 h and was compared with the hierarchical morphokinetic model of Meseguer et al. (2011). (iii) A ConvNeXt-Tiny network with three output heads assigns Gardner grades to day-5 blastocyst images (2,344 images, 851 patients), trained with five-fold patient-level cross-fitting. (iv) The association of expert and automated grades with live birth was analysed in 603 transferred embryos. All partitions created for this work were made at the patient level, and 95% confidence intervals (CIs) were obtained by a patient-cluster bootstrap.

Results. On 141 test videos from 108 held-out patients (58,988 annotated frames), decoded phase accuracy was 0.718 (0.928 within one phase), and the median absolute error of division onsets was 0.20 h for t2, 0.30 h for t3 and 0.50 h for t4, against a frame interval of approximately 0.25 h. The deep model timed t2 to within 1 h in 92% of test videos, compared with 44% for the classical baseline. Rankings computed from automatically detected timings reached an area under the ROC curve (AUC) of 0.690 (95% CI 0.566–0.794), compared with 0.717 (0.611–0.809) when the same model used manual annotations. On the 300-image consensus test set, the five-model grading ensemble reached accuracies of 0.853 (0.81–0.89) for expansion, 0.714 (0.66–0.77) for the inner cell mass and 0.660 (0.60–0.72) for the trophectoderm, against 0.85, 0.69 and 0.62 for the best published models; the evaluation code reproduced the published benchmark values, and a controlled experiment found no evidence that patient overlap between the official training and test sets inflated these results. No predictor discriminated live birth materially better than chance (AUC 0.47–0.56), and none improved significantly on maternal age alone.

Conclusions. Early division times can be extracted automatically with errors close to the imaging interval, and rankings based on them approach those based on manual annotation. Automated Gardner grading reaches the level of the published benchmark. Accurate grading, however, did not translate into prediction of live birth in this cohort, which underlines the distinction between reproducing expert assessment and predicting clinical outcome.

Keywords: in vitro fertilization; time-lapse imaging; morphokinetics; blastocyst grading; deep learning; evaluation methodology

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1 Introduction

Selecting the embryo with the highest developmental potential is a central decision in IVF. In routine practice the decision rests on morphological assessment by embryologists, most commonly with the Gardner and Schoolcraft scale for blastocysts, which scores the expansion of the blastocoel and the quality of the inner cell mass (ICM) and trophectoderm (TE) [1, 2]. Morphological grades are known to vary between observers: in the benchmark dataset used in this work, individual embryologists agreed with a consensus panel on 70–78% of grades, depending on the criterion [11].

Time-lapse imaging (TLI) records embryos at regular intervals without removing them from the incubator and allows the timing of developmental events to be annotated [3]. Such morphokinetic parameters have been associated with blastocyst formation [4] and implantation [5, 6]. Annotation is nevertheless manual, time-consuming and itself variable, and published morphokinetic selection algorithms have shown limited predictive value when applied in clinics other than those in which they were developed [7]. These limitations motivate automated annotation [8, 9] and, more broadly, the use of deep learning for embryo assessment [12–14]. The clinical benefit of such systems remains unproven: a randomized trial of a deep-learning selection system could not demonstrate non-inferiority to morphological assessment for clinical pregnancy [15], and time-lapse-based selection did not improve ongoing pregnancy rates in a large multicentre trial [16]. Retrospective evaluations are additionally susceptible to selection bias, data leakage and non-comparable metrics [17–20].

This report describes **Embryo Scan**, an open pipeline that addresses three tasks on two public datasets (Figure 1): automated annotation of division timings from time-lapse video, ranking of embryos by developmental potential, and Gardner grading of day-5 blastocyst images. Particular attention is given to the validity of the evaluation. The contributions are:

1. **Automated morphokinetic annotation.** A frame-level phase classifier with three-frame temporal context, combined with monotonic dynamic-programming decoding, which times the first divisions of held-out embryos with median errors of 0.2–0.5 h.
2. **A classical baseline under identical scoring.** A training-free MATLAB pipeline (circular Hough localization, motion energy and rules derived from embryology) quantifies the contribution of learning to the timing task.
3. **End-to-end viability ranking.** Rankings computed from automatically detected timings are compared with rankings from manual annotations and with a published hierarchical model.
4. **Gardner grading at benchmark level, with verified evaluation.** The evaluation code reproduces the published benchmark values, and a controlled retraining experiment tests whether patient overlap in the official data split inflates the results.
5. **A negative result on live birth.** Using paired bootstrap comparisons against clinical baselines, neither expert nor automated morphology predicted live birth among transferred embryos.
6. **Open software.** Code, fixed patient-level splits, tests, continuous integration and an interactive web demonstrator are publicly available.

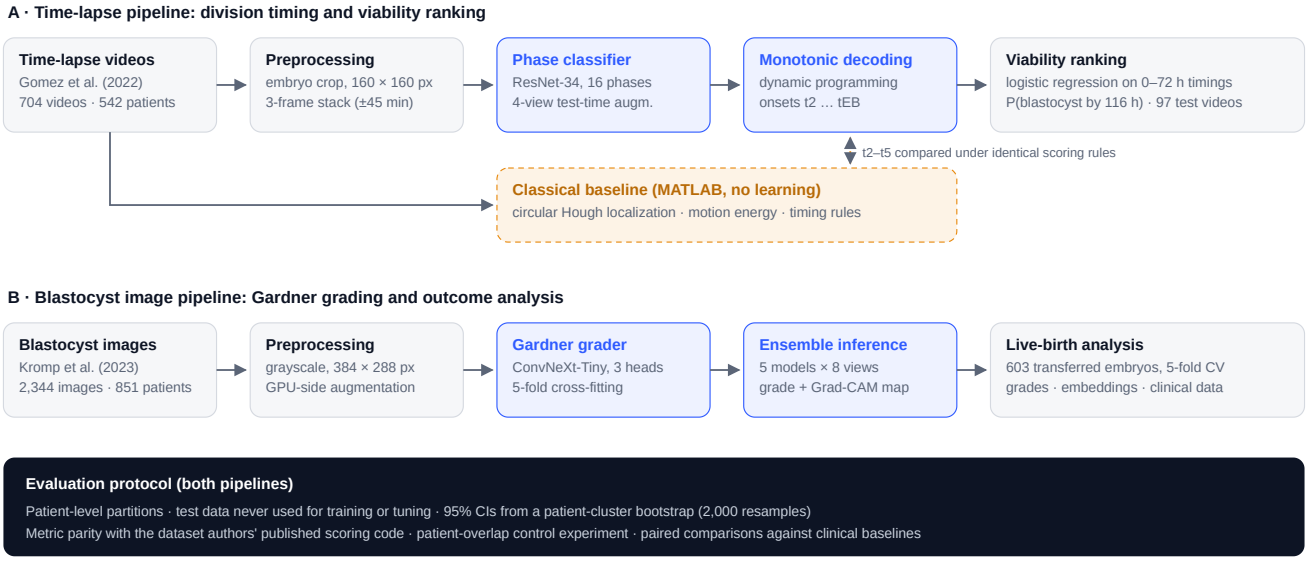


Figure 1. Overview of the analysis. (A) Time-lapse recordings are classified frame by frame, decoded into a monotonic phase sequence and converted into division times, which feed a viability ranking; a classical baseline is scored with the same rules. (B) Day-5 blastocyst images are graded by a five-model ensemble, and the resulting grades and image embeddings are analysed against live birth.

2 Background and Related Work

2.1 Morphological and morphokinetic assessment

The Gardner and Schoolcraft system grades blastocyst expansion on an ordinal scale (1, early blastocyst, to 6, hatched blastocyst) and grades the ICM and the TE from A to C according to the number and cohesion of their cells; the ICM and TE are graded only from expansion grade 3 onwards [1]. Consensus recommendations for embryo assessment were summarized in the Istanbul consensus [2]. Higher blastocyst grades have been associated with implantation and live birth [44, 45].

TLI systems capture images at intervals of a few minutes throughout culture. A standard nomenclature defines the time of each event in hours after insemination, including the appearance and fading of the pronuclei (tPNa, tPNf), the cleavage times t2–t9+, compaction (tM), the start of blastulation (tSB) and the full and expanded blastocyst stages (tB, tEB) [3]. Wong et al. showed that cell-cycle parameters measured before embryonic genome activation predict development to the blastocyst stage [4]. Meseguer et al. proposed a hierarchical classification based on t5, the synchrony of the second division ($s2 = t4 - t3$) and the duration of the second cell cycle ($cc2 = t3 - t2$), with implantation rates decreasing from category A to D [5]. Direct cleavage from one to three cells ($cc2 < 5$ h) has been associated with very low implantation rates [6]. Barrie et al. applied six published selection algorithms to a single external cohort and found none with clinically useful predictive value, and concluded that algorithms should be developed in-house [7].

2.2 Automated annotation of time-lapse sequences

Feyeux et al. developed an automated annotation tool based on the grey-level coefficient of variation and the thickness of the zona pellucida, and reported high overall concordance with manual annotation ($r^2 = 0.92$) on 701 videos [8]. Lau et al. framed embryo staging as frame classification, combined a weakly supervised region-proposal network with a residual classifier and decoded the predictions with a dynamic program that constrains the sequence of stages to be monotonic [9]. The decoding used in the present work follows the same principle. Gomez et al. released the time-lapse dataset used here to support benchmarking of morphokinetic parameter prediction [10].

2.3 Deep learning for embryo assessment

Khosravi et al. trained a convolutional network (STORK) to discriminate good-quality from poor-quality blastocysts from time-lapse images [12]. Tran et al. predicted fetal heart pregnancy directly from time-lapse videos [13], and Berntsen et al. described a fully automated time-lapse model (iDAScore) evaluated on data from 18 clinics [14]. Kromp et al. released the annotated blastocyst dataset used here, together with a benchmark of human experts and published architectures, including Xception [42] and the Swin Transformer [43], for the prediction of Gardner criteria [11]. Reported discrimination depends strongly on the evaluation population. In the study of Berntsen et al., the AUC was 0.95 when all embryos, including discarded ones, were ranked, but 0.67 among transferred embryos with known implantation data [14]. Prospective evidence remains limited: a randomized, double-blind trial could not demonstrate non-inferiority of deep-learning-based selection to standard morphology for clinical pregnancy [15], and the SelecTIMO trial found no improvement in ongoing pregnancy with time-lapse-based selection [16].

2.4 Evaluation pitfalls and reporting standards

Kragh and Karstoft showed that selection bias is unavoidable when AI models are compared with current selection practice in retrospective cohorts, because outcomes are known only for the embryos that were transferred [17]. Afnan et al. argued for interpretable models and randomized evaluation before clinical use [18]. More generally, data leakage between training and test sets is a widespread cause of over-optimistic results in machine-learning-based science [19, 20]. Reporting guidelines for prediction models (TRIPOD+AI) [21] and for medical-imaging AI (CLAIM 2024) [22], together with model cards [23], set out the information needed to assess such studies. Appendix D maps this report to the principal TRIPOD+AI items.

3 Materials

3.1 Time-lapse dataset

The time-lapse dataset of Gomez et al. [10] comprises 704 recordings of embryos from intracytoplasmic sperm injection (ICSI) cycles at a single centre, acquired with an EmbryoScope system at approximately 15-minute intervals for up to six days. Each recording is provided at seven focal planes; only the in-focus reference plane was used in this work (about 311,000 frames). For every video, embryologists annotated the start and end frames of 16 phases (Table 1). Times are expressed in hours after insemination. Patient identity was derived from the video identifier (for example, `ADM715-1-2` → `ADM715`).

Table 1. Developmental phases annotated in the time-lapse dataset [3, 10].

Code	Event	Code	Event
tPB2	extrusion of the second polar body	t7, t8	7 and 8 cells
tPNa	appearance of the pronuclei	t9+	9 or more cells
tPNf	fading of the pronuclei	tM	compacted morula
t2	2 cells	tSB	start of blastulation
t3	3 cells	tB	full blastocyst
t4	4 cells	tEB	expanded blastocyst
t5, t6	5 and 6 cells	tHB	hatching blastocyst

The videos were divided at the patient level, with a fixed random seed, into training (490 videos, 380 patients), validation (73 videos, 54 patients) and test (141 videos, 108 patients) partitions. The partition is committed to the repository (`splits/timelapse-videos.csv`). The validation and test partitions contain 31,770 and 58,988 annotated frames, respectively.

Viability outcome. The video dataset contains no pregnancy outcomes. Blastocyst formation by 116 h (day 5) was therefore used as the endpoint, defined as the first annotated frame at phase tB or later at or before 116 h. To avoid treating a truncated recording as developmental arrest, only videos recorded for at least 116 h were included (486 videos, 58% positive). The ranking model was trained on 389 videos from the training and validation partitions and evaluated on 97 test videos (61 positive).

3.2 Blastocyst image dataset

The dataset of Kromp et al. [11] contains 2,344 images of day-4 and day-5 blastocysts (512 × 384 pixels) from 851 patient identifiers. Of these, 2,042 images carry “silver” labels from a single embryologist, and 300 images form a “gold” test set labelled by the majority vote of a panel of embryologists. One image with two conflicting silver labels and one unlabelled image were excluded. Expansion is graded 1–5; ICM and TE are graded A–C, or recorded as not gradable below expansion grade 3. Clinical data and outcomes are provided for 754 transferred embryos from 732 patients: maternal age, anti-Müllerian hormone (AMH), endometrial thickness, numbers of cumulus–oocyte complexes (COC) and metaphase II oocytes (MII), the day of transfer (day 4, 112; day 5, 642), and three outcomes, namely a positive pregnancy test (42.0%), ongoing pregnancy with fetal heartbeat (34.6%) and live birth (31.0%). Live birth was chosen as the primary outcome. Twenty-two patients underwent double embryo transfer; both embryos inherit the outcome of the transfer, which introduces label noise.

Data curation. Several defects in the released files were identified and corrected (Table 2). The most consequential was the conversion of numeric values into dates by a spreadsheet with German locale settings (for example, the AMH value 1.26 stored as `Jän.26`). Values of the form `DD.Mon` with a month below 10 are ambiguous (`01.Apr` may encode 1.4 or 1.04); these were assigned the midpoint, and a sensitivity analysis compared the two extreme readings. Image names were matched on (patient, embryo) pairs because zero-padding differed between the annotation files and the image files.

Table 2. Data defects corrected before analysis, and missing values (blastocyst dataset).

Defect	Rows
AMH stored as a date, `Mon.YY` form (e.g. `Jän.26` = 1.26)	377
AMH stored as a date, `DD.Mon` form (e.g. `02.Nov` = 2.11)	124
… of which ambiguous (`01.Apr` = 1.4 or 1.04; midpoint used)	118
Endometrial thickness stored as a date (`08.Mai` = 8.5 mm)	144
Oocyte counts written as sums (`10+1`)	9
Image with conflicting duplicate silver labels	1
AMH, endometrium, COC missing	165, 12, 5

Partitions. All partitions created for this work were made at the patient level, because sibling embryos share genetic background and imaging conditions (Table 3). For the outcome task, about 20% of patients were reserved as a locked test set (151 embryos), and the remaining development set (603 embryos, 585 patients, 187 live births) was divided into five folds stratified by outcome. The locked test set was not used in any analysis reported here. For the grading task, the 300 gold images formed the test set, images of patients in the locked outcome test set were held out from all grading models (345 images), and the remaining 1,697 silver-labelled images formed a training pool divided into five patient-level folds stratified by expansion grade. The official gold set was drawn by image rather than by patient, so 511 pool images (30%) belong to 163 patients who also contribute gold images. Section 4.7 describes how the effect of this overlap was measured.

Table 3. Partitions of the blastocyst dataset. All partitions are patient-level except the official gold test set, which was drawn by image (Section 4.7).

Task	Partition	Patients	Images or embryos	Use
Grading	training pool	661	1,697	five-fold cross-fitting
Grading	gold test	249	300	evaluation only
Grading	held out	141	345	never seen by grading models
Outcome	development	585	603 (187 live births)	five-fold cross-validation
Outcome	locked test	147	151 (47 live births)	not used in this report

4 Methods

4.1 Developmental phase classification

Preprocessing. In each video, the embryo was located as the most textured region (edge energy averaged over twelve frames, with the bright rim of the culture well and a border excluded), and a 288 × 288-pixel crop around it was resized to 160 × 160 pixels. Truncated JPEG files were decoded as far as possible, and unreadable frames were flagged and excluded from training.

Input representation. Cell divisions are brief events that a single frame often does not resolve. The input for frame t therefore stacks the grey-scale frames $t - 3$, t and $t + 3$ (approximately 45 minutes apart, clamped at the ends of the video) as three channels, which allows the network to recognise cells in the course of dividing.

Architecture and training. A ResNet-34 [24] initialized with ImageNet weights trained with the procedure of Wightman et al. [25] (`resnet34.a1_in1k`) was fitted with a 16-way output and dropout of 0.2. Training samples were drawn from every third frame of the training videos (the sampling stride equals the stacking offset, so all stacked neighbours lie on the same grid); frames before the first annotation and unreadable frames were excluded. The loss was cross-entropy with label smoothing of 0.05 [29] and class weights proportional to the inverse square root of class frequency. The model was optimized with AdamW [28] (learning rate 6×10^{-4}, weight decay 0.05, batch size 96) for eight epochs, with a linear warm-up over half an epoch followed by cosine decay, in mixed precision. Augmentation was applied on the GPU, with the same transformation for all three channels: rotation by an arbitrary angle (embryos have no natural orientation), horizontal flipping, scaling by a factor of 0.87–1.15, translation of up to 4% of the image extent, perturbation of brightness and contrast, and Gaussian blur with probability 0.2. Validation accuracy was monitored during development, and training ran for a fixed number of epochs without early stopping. At inference, class probabilities were averaged over the four right-angle rotations.

4.2 Monotonic decoding and event timing

Development is irreversible: an embryo cannot return to an earlier phase. Given per-frame log-probabilities $\log p_\theta(c \mid \mathbf{x}_t)$ over phases $c \in \{1, \ldots, C\}$, the decoder finds the most probable non-decreasing phase sequence,

$$\hat{c}_{1:T} = \arg\max_{c_1 \le c_2 \le \cdots \le c_T} \sum_{t=1}^{T} \log p_\theta(c_t \mid \mathbf{x}_t),$$

by dynamic programming over the recurrence

$$B_t(c) = \log p_\theta(c \mid \mathbf{x}_t) + \max_{c' \le c} B_{t-1}(c'),$$

followed by back-tracking. A running maximum gives a cost of $O(TC)$. The decoder corresponds to Viterbi decoding [31, 32] in a left-to-right hidden Markov model with uniform transition scores and follows the monotonic decoding of Lau et al. [9]. Phases may be skipped, for example when two divisions occur between consecutive frames. The onset of phase k is

the time of the first frame whose decoded phase is k or later. When a phase is skipped, its onset therefore coincides with that of the next phase, which matches the definition applied to the annotations.

Metrics. Frame-level accuracy was computed before and after decoding, together with the proportion of frames within one phase of the annotation. For each division event, the error is the detected minus the annotated onset in hours. The median absolute error and the proportions of videos within 1 h and 2 h are reported over the videos in which both the annotation and the prediction reach the event. Agreement of the Meseguer category (Section 4.4) derived from detected and from annotated timings was also recorded.

4.3 Classical baseline

To quantify the contribution of learning, a training-free pipeline was implemented with the MATLAB Image Processing Toolbox (R2024a).

1. **Localization.** The embryo was located by a circular Hough transform (`imfindcircles`, two-stage method [33]) applied to the median of the first 20 frames, with a radius range that excludes the rim of the well. When no circle was found, the peak of smoothed local contrast was used instead.
2. **Motion energy.** For each frame, the mean absolute difference from the preceding frame inside the embryo was computed after Gaussian smoothing and normalization of frame brightness, and was expressed as a robust z-score relative to the video's median and median absolute deviation. Divisions produce spikes because a cleaving cell changes many pixels at once.
3. **Timing rules.** Spikes were defined as local maxima of a three-frame moving average with a prominence of at least 1.5 and a separation of at least 0.5 h. Rules derived from typical embryo timing assign t2 to the strongest spike between 18 and 38 h, t3 to the first spike 5–18 h later, t4 to the next spike within 4 h of t3 (or t3 itself if none occurs, which corresponds to synchronous division) and t5 to the first spike 6–20 h after t4.

The rules were checked on the validation videos, and the pipeline was then applied once to the test videos. The deep model's errors were recomputed from its saved predictions with the same rules: a division that a method failed to detect counts as a failure in the within-1 h and within-2 h proportions, and median errors are taken over detected divisions.

4.4 Viability ranking

Only information available by day 3 was used. Onsets later than 72 h were discarded, and the features were the onsets t2–t8, cc2, s2, an indicator of direct cleavage (cc2 < 5 h) [6] and an indicator of reaching eight cells by 72 h. Missing values were imputed with the training median, with missingness indicators, and the features were standardized before fitting an L2-regularized logistic regression ($C = 0.5$). The model was trained on manual annotations from the training and validation videos and evaluated on the test videos in two settings: with the embryologists' annotations, and end to end, with timings detected by the phase model from frames up to 72 h only. The comparator was the hierarchical model of Meseguer et al. [5]: t5 within 48.8–56.6 h (categories A or B, otherwise C or D), s2 ≤ 0.76 h (A or C, otherwise B or D) and cc2 ≤ 11.9 h (+, otherwise −), mapped to an ordinal score from A+ (8) to D− (1), with 0 for embryos that cannot be categorized. The morphological exclusion criteria of the original model could not be applied because the corresponding annotations are not available.

4.5 Blastocyst grading

Architecture. A ConvNeXt-Tiny backbone [26] (ImageNet-22k pre-training, fine-tuned on ImageNet-1k; `convnext_tiny.fb_in22k_ft_in1k`) produces a 768-dimensional pooled representation. Three linear heads, applied after dropout of 0.3, predict expansion (five classes), ICM (three classes) and TE (three classes). The input is the grey-scale image resized to 384 × 288 pixels and replicated across three channels.

Objective. The loss is the sum of the cross-entropy for expansion and class-weighted cross-entropies for ICM and TE, with weights proportional to the inverse square root of class frequency. The ICM and TE terms are masked where the grade is not defined, and label smoothing of 0.05 was applied.

Optimization. Training used AdamW with learning rates of 1×10^{-4} for the backbone and 1×10^{-3} for the heads (weight decay 0.05, batch size 16) for 25 epochs, with one epoch of linear warm-up followed by cosine decay, gradient-norm clipping at 2.0 and mixed precision. An exponential moving average of the weights (decay 0.998) was used for inference. The number of epochs was fixed, so the held-out fold never influenced training. Augmentation was as in Section 4.1, with corners exposed by rotation filled with the median background grey level.

Cross-fitting and inference. The training pool was divided into five patient-level folds. Each fold model was trained on four folds and predicted the fifth, so that every pool image received an out-of-fold prediction; gold and held-out images were graded by the mean of the five models' probabilities. Probabilities were averaged over the eight dihedral transformations of the image (four rotations, with and without reflection). The grade is the class with the highest probability, and ICM and TE are reported as not gradable when the predicted expansion grade is 1 or 2.

Evaluation. Accuracy was computed exactly as in the dataset authors' scoring script, in which ICM and TE are scored only when neither the model nor the panel assigns expansion grade 1 or 2. To verify the implementation, the published Xception predictions distributed with the dataset were re-scored with the same code. Weighted F1 scores and the quadratically weighted kappa for expansion [37] are also reported. Confidence intervals were obtained by a patient-cluster bootstrap over the gold set.

4.6 Decision thresholds for rare grades

Grade C is rare in the training pool (16 ICM and 47 TE labels among approximately 1,700 images), and the argmax decision rule seldom predicts it. A post-hoc adjustment adds a class-specific offset to the log-probabilities of grades B and C, which is related to logit adjustment for long-tailed classification [30]. The offsets were tuned to maximize balanced accuracy on the out-of-fold predictions for the training pool, without use of the gold set, which gave ICM offsets of (0, 0, 0.9) and TE offsets of (0, 0.1, 1.3) for grades A, B and C.

4.7 Patient-overlap control experiment

Because the official gold set was sampled by image, 30% of the training pool comes from patients who also appear in the test set, and similarity between sibling embryos could inflate test accuracy. Three five-fold ensembles with identical settings were trained: (i) on the whole pool, as reported (1,358 training images per fold); (ii) excluding all patients with gold images (949 images per fold); and (iii) a size-matched control that removes the same number of randomly selected images (949 images per fold). The overlap effect is estimated by (ii) − (iii), which holds the amount of training data constant, and the effect of the reduced data by (iii) − (i). Paired differences were evaluated on identical bootstrap resamples.

4.8 Live-birth prediction

All analyses used patient-level five-fold cross-validation on the development set (603 embryos, 585 patients, 187 live births). The predictor sets were maternal age alone; the embryologist's grade (expansion as an ordinal variable, ICM and TE as indicator variables); clinical data (age, log AMH, endometrial thickness, COC, MII and day of transfer); the automated grade (the 11 class probabilities of the grading model); the automated image embedding (768 dimensions reduced to 16 principal components within each training fold); and combinations of these. Each embryo's image features came from grading models that had not been trained on that image. Logistic regression was used throughout, with gradient-boosted trees (LightGBM [38]) as an alternative for the clinical model. All imputation, scaling and dimensionality reduction were fitted within the training folds. In addition, the entire grading network was fine-tuned end to end for live

birth, with grading retained as an auxiliary task (loss weight 0.5). Discrimination was measured by the AUC, with paired bootstrap differences against age alone and against the embryologist's grade, and calibration by the Brier score [36], for which a constant prediction of the base rate scores 0.214.

4.9 Statistical analysis

All confidence intervals are 95% percentile intervals from a patient-cluster bootstrap with 2,000 resamples [34, 35], in which whole patients are resampled because embryos from the same patient are not independent. Comparisons between models use paired differences on identical resamples. The intervals reflect sampling variability of the test population, not variability between training runs. Individual grading models from different folds differed by up to 0.04 in expansion accuracy.

4.10 Implementation

The pipeline is implemented in Python (PyTorch [39], timm, scikit-learn [40], LightGBM and pytorch-grad-cam) with a MATLAB baseline. Datasets are cached as arrays of 8-bit integers (memory-mapped for the video frames); training data are kept on the GPU and augmented there in batches, which avoids data-loader worker processes. Training the phase model took approximately 80 minutes on a single NVIDIA RTX 2060 GPU, and training the five grading models took approximately 35 minutes. Continuous integration (GitHub Actions) runs linting (ruff), unit tests (pytest; MATLAB unit tests with synthetic images and signals) and a production build of the web application. The interactive demonstrator uses Three.js and Vite. A FastAPI service grades uploaded images with the five-model ensemble and returns Grad-CAM attribution maps [27] computed on the final ConvNeXt stage. On CPU-only hosting, it uses two test-time views for the grade instead of eight, so its output can occasionally differ from the evaluated configuration.

5 Results

5.1 Phase classification and division timing

Decoding improved frame-level accuracy on the test set from 0.697 to 0.718, and 92.8% of frames were assigned to the annotated phase or an adjacent one (Table 4). Figure 2 shows a representative held-out embryo: isolated per-frame errors are removed by the decoder, and the remaining discrepancies are short offsets at phase boundaries.

Table 4. Phase classification and timing on held-out patients.

Partition	Videos	Annotated frames	Frame accuracy (raw)	Frame accuracy (decoded)	Within one phase	Median absolute onset error
Validation	73	31,770	0.684	0.708	0.916	0.85 h
Test	141	58,988	0.697	**0.718**	0.928	1.00 h

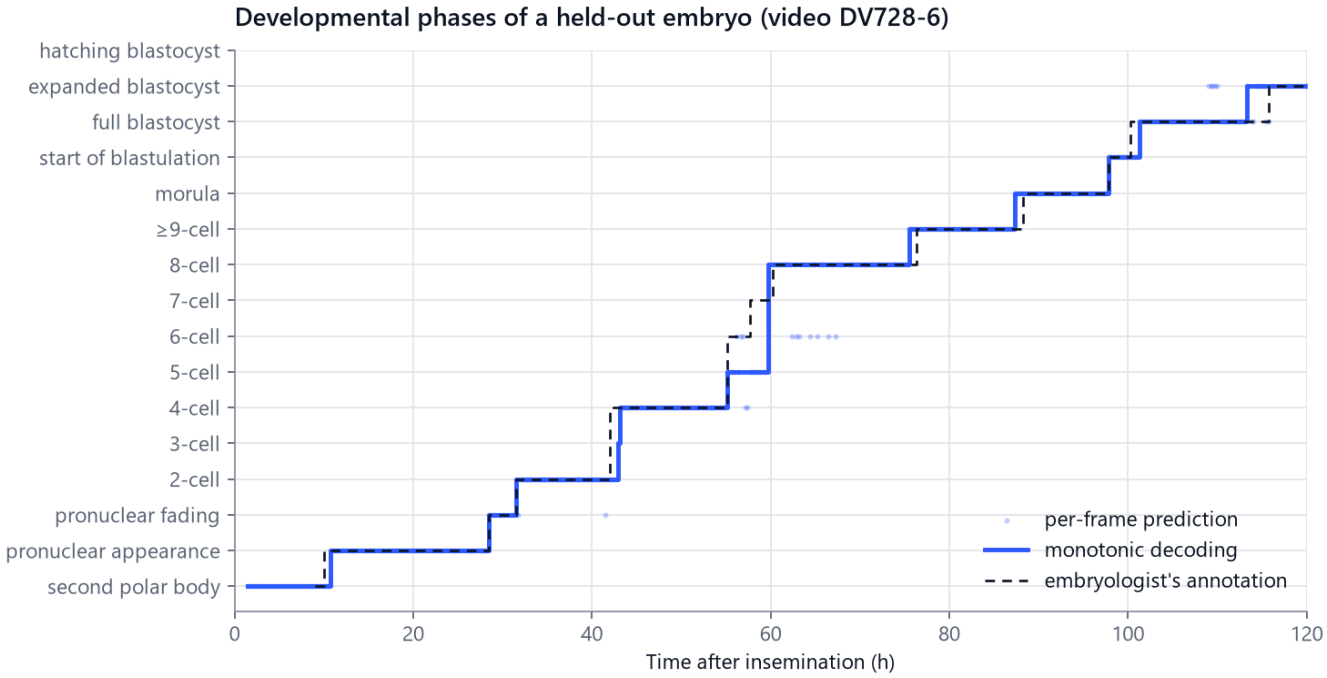


Figure 2. Development of a held-out embryo (video DV728-6). Dots: per-frame predictions; solid line: decoded sequence; dashed line: embryologist's annotation.

Errors increased with developmental stage (Table 5, Figure 3). The first three cleavages were timed to within one or two frame intervals (median absolute error 0.20 h for t2, 0.30 h for t3 and 0.50 h for t4), and 92% of test videos had t2 within 1 h of the annotation. From the eight-cell stage onwards, overlapping blastomeres and gradual transitions (compaction, cavitation, expansion) made the onsets less well defined, and median errors rose to 1.7 h for t8 and tM and to 4.3 h for tEB. The Meseguer category derived from detected timings matched the category derived from annotations in 47% of the 141 test videos. The category depends on sharp thresholds, notably $s2 \leq 0.76$ h, which corresponds to three frame intervals.

Table 5. Timing error per event on the test partition (detected minus annotated onset).

Event	Videos	Median absolute error (h)	Within 1 h	Within 2 h
t2	141	0.20	0.92	0.94
t3	141	0.30	0.78	0.84
t4	141	0.50	0.76	0.87
t5	141	1.00	0.57	0.70
t6	140	0.80	0.59	0.73
t7	137	1.00	0.53	0.69
t8	135	1.70	0.35	0.56
t9+	123	3.10	0.31	0.46
tM	111	1.70	0.40	0.56
tSB	104	1.30	0.48	0.66
tB	94	1.10	0.50	0.68
tEB	73	4.30	0.21	0.29

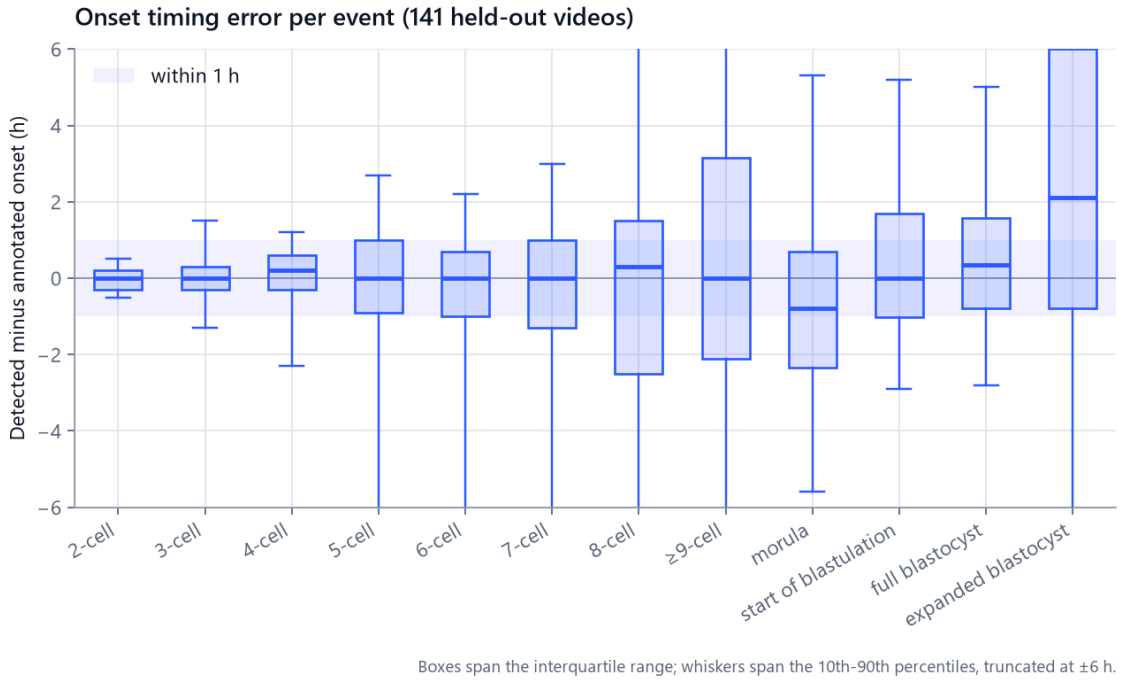


Figure 3. Distribution of timing errors per event on 141 held-out videos. Boxes span the interquartile range; whiskers span the 10th to 90th percentiles and are truncated at ±6 h.

5.2 Comparison with the classical baseline

The circular Hough transform located the embryo in 134 of 141 test videos. Motion-energy spikes coincided with many divisions (Figure 4), but cell movement and drift of the embryo also produced spikes, and the classical pipeline was considerably less accurate than the deep model at every event (Table 6, Figure 5). For t2, 44% of videos were timed within 1 h by the classical pipeline and 92% by the deep model; for t5, the corresponding proportions were 22% and 57%. The Meseguer category matched the annotation-derived category in 25% of 130 categorizable videos with classical timings and in 47% of 141 videos with deep-model timings.

Table 6. Classical baseline and deep model on the same 141 test videos.

Event	Detected (classical)	Median error, classical	Median error, deep	Within 1 h, classical	Within 1 h, deep	Within 2 h, classical	Within 2 h, deep
t2	94%	1.50 h	0.20 h	44%	**92%**	51%	**94%**
t3	92%	3.05 h	0.30 h	34%	**78%**	43%	**84%**
t4	92%	1.30 h	0.50 h	45%	**76%**	50%	**87%**
t5	82%	3.70 h	1.00 h	22%	**57%**	33%	**70%**

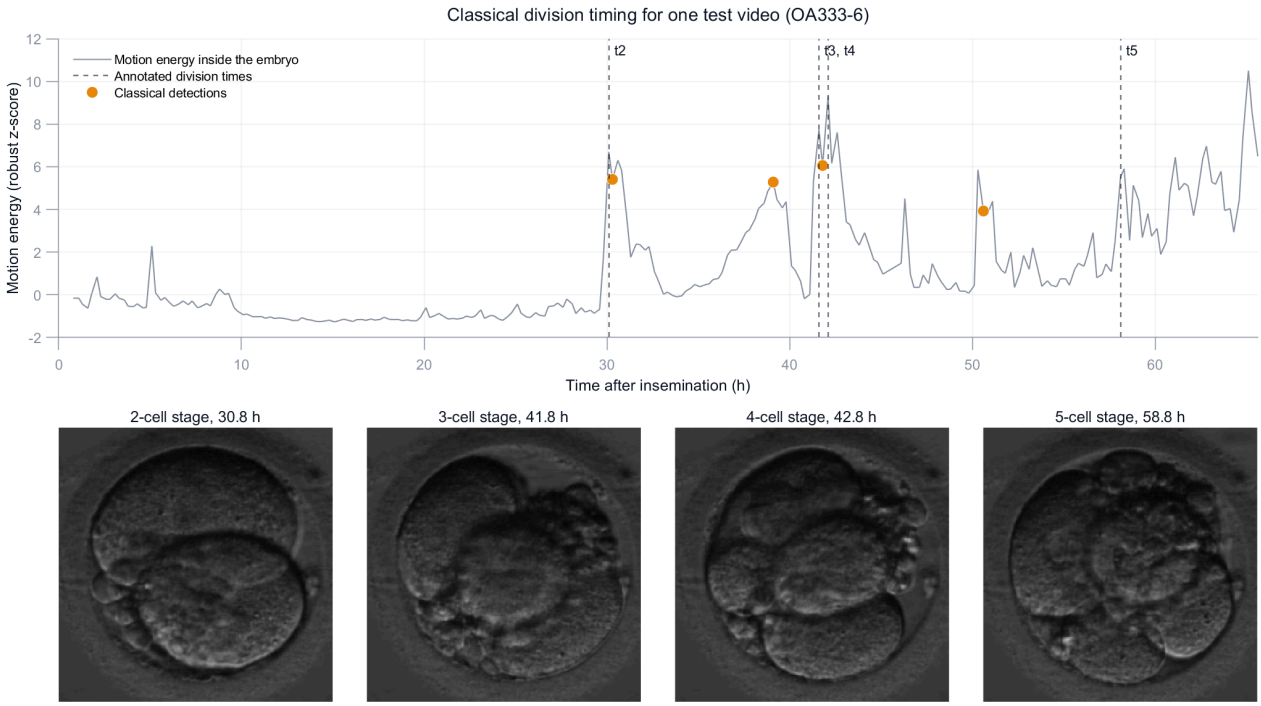


Figure 4. Classical pipeline on one test video (OA333-6). Top: motion energy inside the embryo (robust z-score), with the embryologist's division times (dashed lines) and the classical detections (orange). Bottom: frames at the annotated divisions.

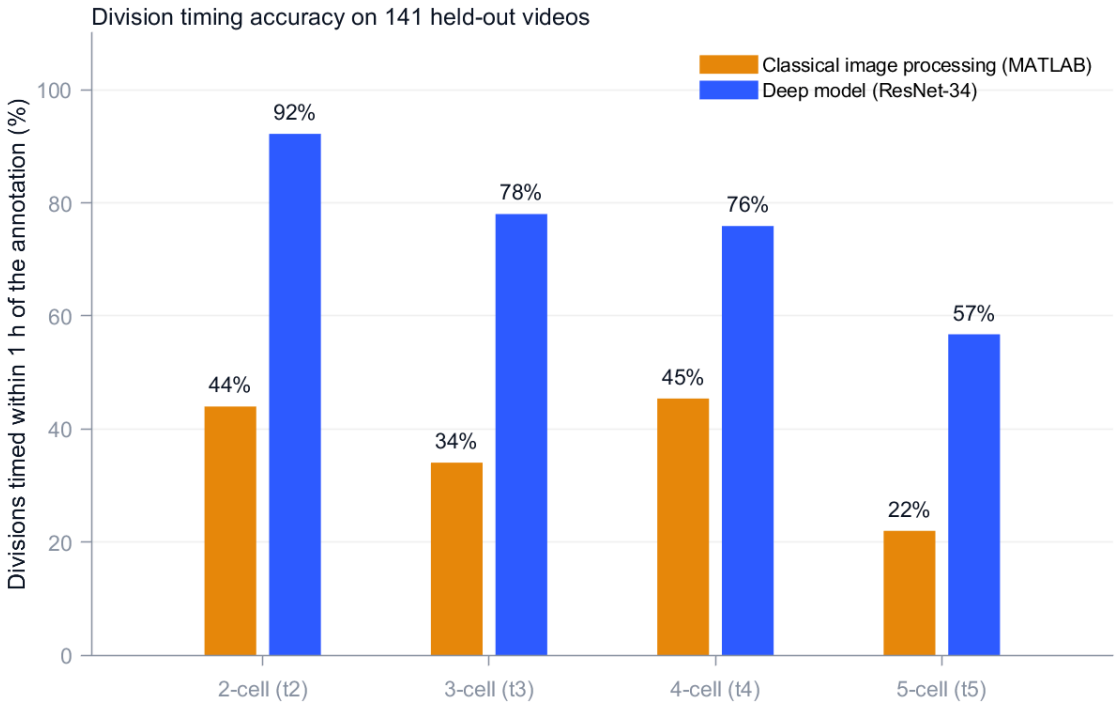


Figure 5. Proportion of the 141 test videos in which each division was timed within 1 h of the annotation.

5.3 Viability ranking

The learned model ranked test embryos by blastocyst formation with an AUC of 0.717 (95% CI 0.611–0.809) when given manual annotations and 0.690 (0.566–0.794) when given automatically detected timings (Table 7, Figure 6). The published Meseguer categories reached 0.675 (0.551–0.779) with annotations and 0.610 (0.480–0.727) with detected timings. The difference of 0.027 between manual and automated inputs to the learned model is small relative to the widths of the intervals (approximately 0.20–0.23); a formal paired test was not performed. The degradation was larger for the categorical

model, which is consistent with its sensitivity to small timing errors near its thresholds. Among test embryos, blastocyst rates did not decrease monotonically across Meseguer categories (Table 8). The categories were developed to predict implantation rather than blastocyst formation, and each category contains only 7–21 embryos.

Table 7. Ranking of 97 test embryos by blastocyst formation by day 5 (61 positive).

Timings from	Ranking method	AUC (95% CI)
Embryologist's annotations	Meseguer et al. 2011 categories	0.675 (0.551–0.779)
Embryologist's annotations	Logistic regression on timings	**0.717** (0.611–0.809)
Phase model, end to end	Meseguer et al. 2011 categories	0.610 (0.480–0.727)
Phase model, end to end	Logistic regression on timings	**0.690** (0.566–0.794)

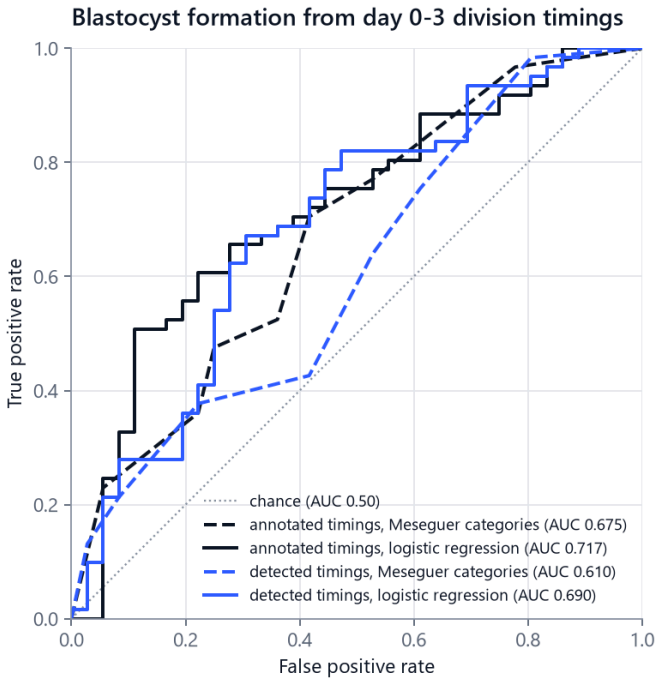


Figure 6. ROC curves for the four rankings in Table 7. Black: timings from annotations; blue: timings detected by the phase model. Solid: logistic regression; dashed: Meseguer categories.

Table 8. Blastocyst formation by day 5 per Meseguer category (annotated timings, test embryos).

Category	A+	A−	B+	B−	C+	C−	D+	D−
Embryos	16	14	8	7	13	8	21	10
Blastocyst rate	0.88	0.57	0.88	0.43	0.85	0.50	0.57	0.20

5.4 Blastocyst grading

Re-scoring the published Xception predictions with the evaluation code gave accuracies of 0.780, 0.686 and 0.616, matching the values reported by the dataset authors (0.78, 0.69 and 0.62). The metric implementation is therefore equivalent to the published benchmark. The five-model ensemble reached accuracies of 0.853 for expansion, 0.714 for ICM and 0.660 for TE (Table 9, Figure 7), with a quadratically weighted kappa of 0.916 for expansion (0.875 for the published Xception predictions). The point estimates are at or above the best published result for each criterion, but the 95% intervals include those values, so the comparison indicates parity rather than demonstrated superiority. The published

comparators are, moreover, the best of several architectures for each criterion. Relative to individual embryologists, the ensemble was more accurate for expansion (0.853 versus 0.78) and less accurate for ICM (0.714 versus 0.74) and TE (0.660 versus 0.70). Ensembling improved on the mean single model for every criterion.

Table 9. Gardner grading accuracy on the 300-image gold test set (panel majority vote). ICM and TE are scored only when neither the model nor the panel assigns expansion grade 1–2.

Grader	Expansion	Inner cell mass	Trophectoderm
Individual embryologists vs. panel [11] (mean ± SD)	0.78 ± 0.12	0.74 ± 0.06	0.70 ± 0.14
Best published model [11]	0.85 (Swin)	0.69 (Xception)	0.62 (Xception / Swin)
Published Xception, re-scored with this code	0.780	0.686	0.616
Embryo Scan, single models (mean, range)	0.831 (0.813–0.853)	0.706 (0.691–0.718)	0.654 (0.647–0.667)
Embryo Scan, five-model ensemble (95% CI)	**0.853** (0.81–0.89)	**0.714** (0.66–0.77)	**0.660** (0.60–0.72)
Ensemble, weighted F1	0.852	0.672	0.608

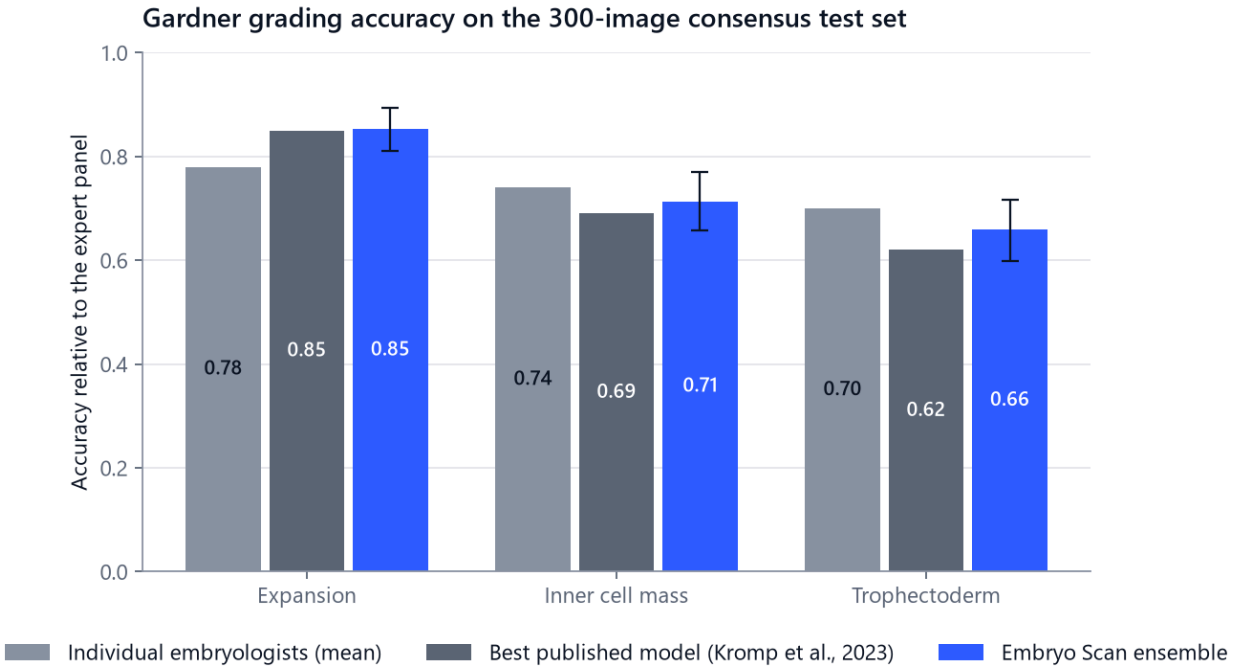


Figure 7. Accuracy against the expert panel. Error bars show the 95% patient-bootstrap interval of the ensemble.

Error structure. Expansion errors were almost exclusively between adjacent grades (Appendix B). The dominant ICM error was the assignment of grade A to embryos that the panel graded B (35 of 68 such embryos). The training labels come from a single embryologist, so systematic differences between that annotator and the panel are learned by the model. The rare grade C was seldom predicted under the argmax rule (ICM 1 of 16; TE 3 of 30). The balanced thresholds of Section 4.6 increased the number of TE grade-C embryos identified from 3 to 12 of 30 and raised the TE macro F1 from 0.525 to 0.615, at the cost of a small decrease in ICM accuracy (Table 10).

Table 10. Effect of balanced decision thresholds (tuned on out-of-fold training predictions only).

Decision rule	ICM accuracy	ICM macro F1	ICM grade C found	TE accuracy	TE macro F1	TE grade C found
Argmax	0.714	0.488	1 / 16	0.660	0.525	3 / 30
Balanced thresholds	0.697	0.489	2 / 16	0.672	0.615	12 / 30

5.5 Effect of patient overlap

Removing all training images from patients with gold images did not reduce accuracy relative to the size-matched control (Table 11). The estimated overlap effects were −0.007 for expansion, +0.008 for ICM and +0.004 for TE, and all 95% intervals included zero. The effect of reducing the training data by 30% was similarly small. Within the reported model, gold images from patients represented in the training pool were graded more accurately for expansion and TE but less accurately for ICM than images from patients absent from it (0.858, 0.697 and 0.684 versus 0.845, 0.744 and 0.616). These subgroups differ in composition, so the comparison is less informative than the controlled experiment above. Overall, the results provide no evidence that the image-level split of the official benchmark inflates the reported accuracies.

Table 11. Patient-overlap control. Accuracy (95% CI) of three five-fold ensembles on the gold set, and paired differences.

Model	Training images per fold	Expansion	Inner cell mass	Trophectoderm
Whole pool (reported)	1,358	0.853 (0.81–0.89)	0.714 (0.66–0.77)	0.660 (0.60–0.72)
Without gold patients	949	0.847 (0.80–0.88)	0.733 (0.68–0.79)	0.650 (0.59–0.71)
Size-matched control	949	0.853 (0.81–0.89)	0.724 (0.67–0.78)	0.646 (0.59–0.70)
Overlap effect (without gold patients − control)		−0.007 (−0.030 to +0.016)	+0.008 (−0.016 to +0.034)	+0.004 (−0.028 to +0.035)
Data-quantity effect (control − whole pool)		+0.000 (−0.026 to +0.026)	+0.011 (−0.013 to +0.036)	−0.014 (−0.041 to +0.013)

5.6 Live-birth prediction

Live birth was poorly predicted by every model (Table 12). Maternal age alone gave an AUC of 0.533 (0.482–0.584), and the embryologist's grade gave 0.550 (0.497–0.598). The best model, combining the automated grade, the embryologist's grade and age, reached 0.564 (0.514–0.612), and no model improved significantly on age alone. The only paired difference whose interval excluded zero was negative: logistic regression on all clinical variables performed worse than age alone (−0.066, −0.115 to −0.017), which suggests that the additional variables contributed mainly noise. Brier scores were indistinguishable from that of a constant prediction of the base rate (0.214). Fine-tuning the entire grading network for live birth gave an AUC of 0.54. Among all 754 transferred embryos, univariate live-birth rates showed the expected direction only at the extremes and with few embryos: 0.08 for women older than 40 (24 embryos) and 0.07 for TE grade C (15 embryos), against 0.31 overall. The choice of reading for ambiguous AMH values did not affect the clinical model (AUC 0.469 versus 0.466).

Table 12. Discrimination of live birth among 603 transferred embryos (patient-level five-fold cross-validation; paired bootstrap differences).

Predictors	AUC (95% CI)	Difference vs. age alone	Difference vs. embryologist's grade	Brier score
Age alone	0.533 (0.482–0.584)	—	−0.017 (−0.085 to +0.050)	0.213
Embryologist's grade	0.550 (0.497–0.598)	+0.017 (−0.050 to +0.085)	—	0.213
All clinical data	0.467 (0.416–0.518)	−0.066 (−0.115 to −0.017)		0.221
All clinical data (LightGBM)	0.478 (0.423–0.532)	−0.055 (−0.124 to +0.014)		0.231
Embryologist's grade + clinical	0.498 (0.445–0.548)	−0.035 (−0.095 to +0.020)		0.221
Automated grade	0.557 (0.508–0.605)	+0.024 (−0.041 to +0.091)	+0.008 (−0.040 to +0.056)	0.214
Automated image embedding	0.518 (0.470–0.566)	−0.015 (−0.078 to +0.055)	−0.032 (−0.088 to +0.025)	0.218
Automated grade + clinical	0.530 (0.481–0.577)	−0.003 (−0.058 to +0.055)	−0.019 (−0.070 to +0.032)	0.220
Automated embedding + clinical	0.502 (0.452–0.552)	−0.031 (−0.089 to +0.032)	−0.048 (−0.101 to +0.009)	0.223
Automated grade + embryologist's grade + age	0.564 (0.514–0.612)	+0.031 (−0.025 to +0.089)	+0.015 (−0.030 to +0.059)	0.213

6 Discussion

6.1 Principal findings

Three findings stand out. First, the timing of early cleavage divisions, which carries much of the morphokinetic information used in published selection models, can be extracted from time-lapse video with median errors of one to two imaging intervals. The learned model is substantially more accurate than a carefully designed classical pipeline under identical scoring. Second, a ranking computed from automatically detected timings performs close to the same ranking computed from manual annotations. A fully automated morphokinetic workflow therefore appears feasible for this endpoint, although the confidence intervals are wide. Third, a modern convolutional network trained on single-annotator labels grades blastocysts at the level of the published benchmark, with an evaluation whose correctness and robustness to patient overlap were tested explicitly. Neither the expert grade nor any automated representation, however, predicted live birth among transferred embryos.

6.2 Interpretation

Temporal context and structural priors. The three-frame input lets the classifier detect cells in the course of dividing, and the monotonic decoder removes physically impossible regressions. This prior is cheap and transparent and corrects isolated errors (Figure 2). The decoder cannot recover a systematic offset, however, and the increase in error after the eight-cell stage reflects genuine ambiguity: blastomeres overlap within a single focal plane, and compaction, cavitation and expansion are gradual processes whose onset is defined by convention. Because each video carries a single annotation, inter-annotator variability could not be measured, and it is unknown how much of the residual error at these stages reflects ambiguity in the reference rather than error of the model.

Why the classical baseline fails. Motion energy responds to every change in the image, including cell rearrangement, fragmentation and drift of the embryo, whereas the network learns the appearance of a division. The rules also propagate errors: a mistimed t2 shifts the search windows for t3–t5. The baseline is nevertheless informative. It shows that a

substantial share of the problem is not solvable with generic signal processing, and it provides an interpretable reference against which the deep model's improvement can be measured.

Thresholded clinical rules amplify small errors. The Meseguer categories lost more discrimination than the logistic regression when computed from detected timings (a decrease in AUC of 0.065 versus 0.027). A synchrony threshold of 0.76 h spans three frame intervals, so an onset error of one frame can change the category. Continuous models degrade more gracefully, and the same consideration applies to any deployment of published rule-based algorithms on automatically annotated data.

Grading versus outcome. The automated grade reproduces the panel's assessment at least as well as an average individual embryologist for expansion and somewhat less well for ICM and TE. That neither the automated nor the expert grade predicted live birth is consistent with the literature rather than at odds with it. Outcomes are available only for embryos that were transferred, which are typically the best-graded embryos of each cohort; this restricts the range of morphology and attenuates any association [17]. Grades of transferred embryos were concentrated at expansion grade 4 with ICM and TE grade A, live birth also depends on maternal and uterine factors that are not visible in the image, and double embryo transfers add label noise. In a large multicentre study, the AUC for implantation was 0.67 among transferred embryos with known outcomes, against 0.95 when all embryos were ranked [14], and randomized trials have not shown clinical benefit [15, 16]. The present negative result, obtained with paired comparisons against clinical baselines, is a useful calibration point for claims of outcome prediction from single static images.

6.3 Limitations

1. **Single-centre data.** Each dataset comes from one clinic and one imaging system. Performance under other microscopes, culture conditions, patient populations and annotation conventions is unknown, and the poor transferability of published morphokinetic algorithms [7] suggests caution.
2. **Surrogate endpoint for viability.** The video dataset contains no pregnancy outcomes, so viability was defined as blastocyst formation by 116 h. This is a developmental endpoint, not a clinical one.
3. **Sample size.** The test partitions are modest (97 videos for viability, 300 images for grading), and the confidence intervals are correspondingly wide. Differences of a few points between methods are not resolved.
4. **Label quality.** Training labels for grading come from a single embryologist, and the time-lapse annotations come from one annotation per video, so inter-observer variability could not be modelled or measured.
5. **Distribution shift in the ranking model.** The viability model was trained on manual annotations and applied to detected timings. Training on cross-fitted detected timings would align the two distributions and may close part of the remaining gap.
6. **Single focal plane.** Only the in-focus reference plane was used. The additional planes may resolve overlapping blastomeres at later stages.
7. **Interpretability.** Grad-CAM maps are coarse (a 9 × 12 feature grid, smoothed for display) and computed from one model of the ensemble. They indicate image regions that influence a prediction; they do not explain the decision.
8. **Variance between runs.** Each configuration was trained once (five folds for grading), and the reported intervals do not capture variability between training runs.

6.4 Threats to validity

Several safeguards were applied: patient-level partitions for every partition created in this work, with the official grading test set checked separately for patient overlap; test partitions that were never used for training, tuning or threshold selection (thresholds were tuned on out-of-fold training predictions, and the locked outcome test set was not used); preprocessing fitted within cross-validation folds; image features for outcome models derived only from models that had not seen the

image; verification of the metric against the published benchmark; and an explicit control for patient overlap. A residual risk is that development decisions were informed by validation results on the same single-centre data, so external validation remains essential.

7 Conclusions and Recommendations

Automated annotation of time-lapse recordings recovered the early division times that underlie morphokinetic assessment with errors close to the imaging interval, and supported a viability ranking approaching that obtained from manual annotation. Automated Gardner grading matched the published benchmark under a verified evaluation. Neither expert nor automated morphology predicted live birth among transferred embryos in this cohort.

The following steps are recommended for further work:

1. **External validation** on data from other clinics and imaging systems, ideally with multiple annotators per embryo, before any claim of generalization.
2. **Training the ranking model on detected timings** (cross-fitted on the training videos), and **calibration analysis** of the predicted probabilities.
3. **Richer temporal and optical inputs.** Sequence models over longer windows and multi-focal inputs, aimed at the post-eight-cell stages, where errors are largest.
4. **Modelling of label uncertainty.** Use of multi-rater labels, or explicit modelling of annotator bias, to address the systematic ICM B → A error.
5. **Outcome studies designed for the question.** Outcome prediction should be evaluated on cohorts in which the selection mechanism is known, reported according to TRIPOD+AI [21], and ultimately assessed in randomized trials [15, 16]. The locked outcome test set should be used only for confirmatory evaluation of a pre-specified model.

8 Reproducibility, Data Availability and Ethics

Code. All code is available at github.com/cha205/embryo-scan under the MIT licence. Each step of each pipeline is a single command (`scripts/`), and every table and figure in this report is regenerated by those scripts into `reports/`. The patient-level partitions are committed (`splits/`), random seeds are fixed, and the commands for reproducing the analysis are listed in the repository README.

Models. The weights of the five grading models are published on the Hugging Face Hub (Cfish14/embryo-scan-grader) with a model card (`deploy/model-card.md`).

Data. Both datasets are public: the time-lapse dataset of Gomez et al. [10] on Zenodo (CC BY-NC-SA 4.0) and the blastocyst dataset of Kromp et al. [11] on figshare (CC BY 4.0). Derived assets distributed with the web application retain these licences.

Ethics. This work is a secondary analysis of publicly released, de-identified datasets. No new data were collected from human participants. The software is intended for research and education only and is not a medical device.

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Appendix A: Training Configurations

Setting	Phase classifier	Gardner grader
Backbone	ResNet-34 (`resnet34.a1_in1k`)	ConvNeXt-Tiny (`convnext_tiny.fb_in22k_ft_in1k`)
Input	3 × 160 × 160 (frames $t-3$, t, $t+3$)	3 × 288 × 384 (grey-scale, replicated)
Outputs	16 phases	expansion (5), ICM (3), TE (3)
Dropout	0.2	0.3
Loss	weighted cross-entropy, label smoothing 0.05	sum of three cross-entropies (ICM/TE weighted and masked), label smoothing 0.05
Class weights	inverse square-root frequency, mean 1	inverse square-root frequency, mean 1 (ICM, TE)
Optimizer	AdamW, lr 6×10^{-4}, weight decay 0.05	AdamW, lr 1×10^{-4} (heads 1×10^{-3}), weight decay 0.05
Schedule	0.5-epoch warm-up, cosine decay	1-epoch warm-up, cosine decay
Epochs, batch size	8, 96	25, 16
Other	mixed precision; every third frame sampled	mixed precision; EMA 0.998; gradient clipping 2.0
Test-time augmentation	4 rotations	8 dihedral views × 5 fold models
Training time	~80 min (RTX 2060)	~35 min for five folds (RTX 2060)

Appendix B: Confusion Matrices for Grading (Gold Test Set, Ensemble)

Rows give the panel grade and columns the model grade; n/d = not gradable, n/a = not assessable by the panel.

Expansion

Panel \ Model	1	2	3	4	5
1	20	3	0	0	0
2	3	17	11	0	0
3	0	5	74	7	0
4	0	0	8	141	4
5	0	0	1	0	4
n/a	0	0	0	0	2

Inner cell mass

Panel \ Model	A	B	C	n/d
A	142	8	1	2
B	35	29	2	2
C	4	11	1	1
n/d	5	5	1	43
n/a	2	6	0	0

Trophectoderm

Panel \ Model	A	B	C	n/d
A	114	8	0	0
B	35	42	0	5
C	0	27	3	0
n/d	4	7	0	43
n/a	4	8	0	0

Appendix C: Validation-Partition Results

Phase timing (73 validation videos). Decoded frame accuracy 0.708 (raw 0.684; 0.916 within one phase); median absolute onset error 0.85 h; Meseguer category agreement 47%.

Event	t2	t3	t4	t5	t6	t7	t8	t9+	tM	tSB	tB	tEB
Median absolute error (h)	0.20	0.50	0.50	0.75	0.60	1.50	1.30	2.90	2.00	1.00	1.00	2.00
Within 1 h	0.96	0.64	0.73	0.64	0.67	0.45	0.44	0.28	0.24	0.53	0.53	0.41

Classical baseline (validation). Embryo located by the Hough transform in 67 of 73 videos; proportion within 1 h (classical vs. deep): t2 51% vs. 96%, t3 41% vs. 64%, t4 45% vs. 73%, t5 24% vs. 64%.

Appendix D: Correspondence with TRIPOD+AI

The table maps the principal TRIPOD+AI reporting items [21] to the sections of this report.

TRIPOD+AI item	Location in this report
Title and structured abstract	Title; Abstract
Background, rationale and objectives	Sections 1, 2
Data sources, setting and eligibility	Sections 3.1, 3.2
Outcome definition and timing	Section 3.1 (blastocyst by 116 h); Section 3.2 (live birth)
Predictors and their measurement	Sections 4.1–4.5, 4.8
Sample size and missing data	Tables 2–3; Section 4.4 (imputation)
Data preparation	Sections 3.2, 4.1, 4.5
Analytical methods and model specification	Sections 4.1–4.8; Appendix A
Performance measures and uncertainty	Sections 4.2, 4.5, 4.9
Class imbalance	Sections 4.1, 4.5, 4.6
Model updating and evaluation strategy	Sections 3.2 (partitions), 4.5 (cross-fitting), 4.7
Participant flow and characteristics	Section 3; Tables 1–3
Model performance	Section 5; Tables 4–12
Interpretation, limitations and implications	Section 6
Code, data and model availability	Section 8
Ethics	Section 8